

Synaptic Memory Lab | One-Page Concept Summary

DATAFORGE 2026 (PATHWAY TRACK: EXPLAIN THE FRONTIER) • NEURIPS 2026 EDUCATION ALIGNMENT

Topic: Synaptic Plasticity as Short-Term Memory

Target Audience: AI Researchers, Engineers & Evaluators

Subject Architecture: Dragon Hatchling (BDH) & BDH-CQ

Falsifiable Claim: Fixed synaptic matrices eliminate KV cache growth; forget via cross-talk

1. Paradigm Shift: From O(N) Expanding KV Cache to O(1) Synaptic Plasticity

Standard Transformer architectures maintain sequence context by appending Key-Value pairs to an expanding KV cache. This causes an $O(N)$ memory footprint per sequence, requiring multi-terabyte distributed GPU VRAM for 128k+ contexts. **Dragon Hatchling (BDH)** maps attention into an evolving biological connectivity matrix $\mathbf{S}_t \in \mathbb{R}^{d \times d}$. Working memory remains strictly constant regardless of sequence length, shifting the hardware bottleneck from memory capacity to compute throughput.

2. Mathematical Mechanism: Local Hebbian Outer-Product Writes & Sparse Read

Given Key (K_t), Value (V_t), and Query (Q_t) projections at token step t :

• **Hebbian Synaptic Write:** $S_t = \lambda S_{\{t-1\}} + \eta (V_t K_t^T)$, with retention decay $\lambda \in (0,1]$ and plasticity rate η .

• **Associative Synaptic Read:** $V_{hat}_t = S_{\{t-1\}} Q_t$ | **Sparse Projection:** $H_t = ReLU(W_x X_t + W_s V_{hat}_t + b)$

Cross-Talk Suppression: BDH enforces ~5% non-negative ReLU activation sparsity. Coordinate collision probability between random keys drops to $(0.05)^2 = 0.0025$, guaranteeing quasi-orthogonality $E[K_i^T K_j] \approx 0$ in the positive orthant and preventing destructive superposition interference.

3. BDH-CQ: Test-Time Reasoning Without Chain-of-Thought or Backpropagation

Unlike Test-Time Training (TTT) which executes gradient descent on evaluation samples, or CoT LLMs which emit thousands of verbalized tokens, **BDH-CQ** solves abstraction benchmarks (ARC-AGI) entirely within latent recurrent space. Demonstration pairs are sequentially integrated into the recurrent synaptic state in $O(1)$ steps with zero backpropagation and zero CoT tokens (**Verified Published Anchor:** 29.5% ARC-AGI accuracy at \$0.00070/task).

4. Architectural Comparison & Illustrative Scaling Landscape

Property	Standard Transformer	State-Space (Mamba)	BDH / BDH-CQ (Synaptic)
Memory Scaling	O(N) Expanding KV Cache (9.6 TB @ 600B)	O(1) Recurrent Hidden State	O(1) Fixed Synaptic State (~225 GB @ 600B)*
Memory Multiplier	1.0x Baseline	Fixed state compression	15.0x (1B) → 42.7x (600B) at 128k context*
Update Operation	Concatenate Key-Value vectors	Linear continuous SSM (A, B, C)	Hebbian Outer-Product ($V_t K_t^T$)
Activation Budget	Dense Softmax Attention	Dense Continuous State	Sparse Non-Negative ReLU (~5% Active)
Crossover Point (N*)	N/A (scales linearly)	Immediate state compression	N* = 256 tokens (KV RAM exceeds Synaptic RAM)

5. Empirical Proofs, Stated Bounds & Evidence Discipline

- **Verified Correctness:** 34 automated unit tests validate closed-form Hebbian updates, Frobenius bounds, and crossover math.
- **Stated Substrate Bounds:** Client engine explicitly bounded at $d \leq 256$ and $N \leq 2000$ to guarantee 60 FPS in-browser simulation without latency.
- **Grounded Anchor & Projections:** ARC-AGI module is anchored to Pathway's published 29.5% / \$0.00070 baseline; memory scaling curves are illustrative pedagogical models.
- **Pedagogical Heatmap:** Real-time step-by-step write animation visually isolates newly activated synapses and displays live Hebbian formula updates.

6. Independently Verified Primary Citations (2021–2026)

1. Yang et al. (2024). *Gated Linear Attention Transformers with Hardware-Efficient Training*. ICML 2024. **PMLR 235:56501–56523 (yang24ab)** / arXiv:2312.06635
2. Schlag, Irie & Schmidhuber (2021). *Linear Transformers Are Secretly Fast Weight Programmers*. ICML 2021. **PMLR 139:9355–9366 (schlag21a)**
3. Sun et al. (2023). *Retentive Network: A Successor to Transformer for Large Language Models*. Microsoft Research. **arXiv:2307.08621 [cs.CL]**
4. Behrouz, Zhong & Mirrokni (2025). *Titans: Learning to Memorize at Test Time*. Google Research. **arXiv:2501.00663 [cs.LG]**
5. Pathway Research (2025/2026). *From Attention to Synapses: Deriving BDH • BDH-CQ Report* (29.5% ARC-AGI Anchor). pathway.com